

- Range: It is set of images of all the elements of A.
Range $\neq$ Co-domain (Range need not to be equal to co-domain)
- If f_1, f_2 are two functions
 $f_1(x) + f_2(x)$ is denoted as $(f_1 + f_2)(x)$
 $f_1(x) \cdot f_2(x)$ is denoted as $(f_1 \cdot f_2)(x)$
 $\frac{1}{f(x)}$ is denoted as $\frac{1}{f}(x)$, ($\frac{1}{f}$ is not equal to inverse function)
- If A and B are two sets with
 $|A| = n$ $|B| = m$ then
number of functions possible from A to B = m^n

3.6 Types of functions:

3.6.1 One - One function (Injection):

$f : A \rightarrow B$ is one to one

iff

$$\forall a \forall b (f(a) = f(b) \Rightarrow a = b)$$

- If $f : A \rightarrow B$ is 1-1 then $|A| \leq |B|$
- If $|A| = m$ and $|B| = n$
no of 1 - 1 function possible from A to B = ${}_n P_m$
- If $m = n = k$, then number of 1:1 function = $k!$

3.6.2 Onto function (Surjection):

$f : A \rightarrow B$ is onto iff

$$\forall b \in B \quad \exists a \in A \text{ such that } f(a) = b$$

- If $f : A \rightarrow B$ is onto then $|A| \geq |B|$
- If $|A| = m$ $|B| = n$ then

$$\therefore \text{ number of onto functions} = \sum_{i=0}^n (-1)^i {}^n C_i (n-i)^m$$

- If $m = n = k$, then number of 1:1 function = $k!$

3.6.3 One-to-One Correspondence (or) Bijection:

A function $f : A \rightarrow B$ is Bijection iff

f is both one-one and onto

- If $f : A \rightarrow B$ is bijection, then $|A| = |B|$
- If $f : A \rightarrow B$ is 1 - 1 and $|A| = |B|$
then f is a bijection
- If $f : A \rightarrow B$ is onto and $|A| = |B|$
then f is a bijection

- If $|A| = |B| = n$ then no of bijections possible from A to B are $n!$
- If $f : A \rightarrow A$ is a function and A is a finite set then
A is 1-1 $\Leftrightarrow$ A is onto

3.6.4 Inverse of a function:

$f : A \rightarrow B$ is invertible if its inverse relation f^{-1} is a function from B to A.

$f : A \rightarrow B$ is invertible $\Leftrightarrow f$ is a bijection

- Inverse doesn't exist if a function is not bijection. However, we can find inverse image of subset of codomain. If $f : A \rightarrow B$ is a function and $S \subseteq B$

Inverse image of $S = \{a \in A \mid f(a) \in S\}$

- If f is a function from A to B and let S, T be subsets of A.

$$f(S \cup T) = f(S) \cup f(T)$$

$$f(S \cap T) = f(S) \cap f(T)$$

However, $f(S \cap T) = f(S) \cap f(T)$ if f is a bijection

- If f is a function from A to B and let S, T be subsets of B

$$f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$$

$$f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$$

$$f^{-1}(\bar{S}) = \overline{f^{-1}(S)}$$

3.6.5 Identity function:

A mapping $I_A : A \rightarrow A$ is called an identity function if

$$I_A = \{(x, x) \mid x \in A\}$$

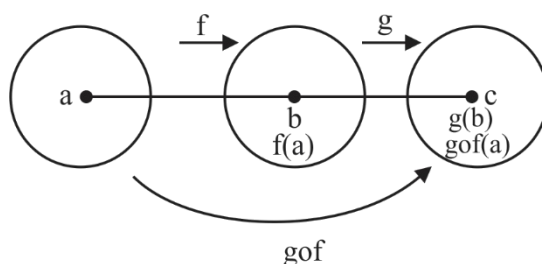
3.6.7 Constant function:

A function $f : A \rightarrow B$ is said to be a constant function if

$$f(x) = c \quad \forall x \in A$$

3.6.8 Composition of functions:

If $f : A \rightarrow B$ and $g : B \rightarrow C$ are two function then $g \circ f : A \rightarrow C$ is called a composition function of f & g.



$$g \circ f(x) = g(f(x))$$

- In $g \circ f : A \rightarrow C$
A is domain of $g \circ f$
C is codomain of $g \circ f$